

Week-6 (DAY-28,29) Hands-On – Keerthivasan R

Problem: Contact Management Web Application

Scenario:

ABC Infotech wants to develop a mini web application using ASP.NET Core MVC to manage contact persons of their clients. The system should allow users to add new contacts, view all contacts, and search for a specific contact by ID. The goal is to build a simple, user-friendly application without using a database, suitable for initial development and learning purposes.

Requirements:

1.Add Model class **ContactInfo** with auto implemented properties:

Name	Type	Modifier
ContactId	int	Public
FirstName	string	Public
LastName	string	public
CompanyName	string	public
EmailId	string	public
MobiliNo	long	public
Designation	string	public

2.Add **ContactController** class and declare variable and write action methods given below.

Declare variable- **static List<ContactInfo>** to collect and maintain contact details.

Functions	Description
public ActionResult ShowContacts()	This action methods send object of type List<ContactInfo> to view and displays contact details in tabular format on view.
public ActionResult GetContactById (int id)	This action method takes id as input and search contact detail using ContactId and send that object of type ContactInfo to view to show contact details.
public ActionResult AddContact()	This action method returns a view to accept contact details from user to add new contact into List<ContactInfo>.
[HttpPost] public ActionResult AddContact (ContactInfo contactInfo)	This action invokes when user click Save button added into above view after entered new contact details. Add new contact details object into a

	List<ContactInfo> and show all the contact in tabular format by redirecting to ShowContacts.
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3. Set ContactInfoController and ShowContacts action method on start-up and run the project.

Technical Constraints

- Use ASP.NET Core MVC (.NET 8 or above)
- Follow MVC Architecture (Model-View-Controller)
- Use strongly typed views
- Use Model Binding for POST requests
- Use LINQ for searching data
- Do NOT use any database (in-memory list only)

Expectations

- Application should run successfully without errors
- Users should be able to:
 - Add new contact
 - View all contacts
 - Search contact by ID
- Proper separation of concerns (MVC pattern)
- Clean and user-friendly UI
- Proper data flow between Controller and View

Learning Outcome

- Understand **ASP.NET Core MVC architecture**
- Learn how to create and use **models, views, and controllers**
- Understand **form handling (GET & POST)**
- Learn **model binding and data passing**
- Practice **LINQ for data filtering**
- Understand **routing and application flow**

- Build a **complete mini end-to-end MVC application**

CODE:

Models/ ContactInfo.cs

```
using System.ComponentModel.DataAnnotations;

namespace ContactApp.Models
{
    public class ContactInfo
    {
        [Required(ErrorMessage = "Contact ID is required")]
        public int ContactId { get; set; }

        [Required(ErrorMessage = "First Name cannot be empty")]
        public string FirstName { get; set; }

        [Required(ErrorMessage = "Last Name is required")]
        public string LastName { get; set; }

        public string CompanyName { get; set; } // Optional field

        [Required(ErrorMessage = "Email is required")]
        [EmailAddress(ErrorMessage = "Invalid Email Address")]
        public string EmailId { get; set; }

        [Required(ErrorMessage = "Mobile Number is required")]
        public long MobileNo { get; set; }

        public string Designation { get; set; }
    }
}
```

Controllers/ ContactController.cs

```
using ContactApp.Models;
using Microsoft.AspNetCore.Mvc;

namespace ContactApp.Controllers
{
    public class ContactController : Controller
    {
        // In memory list
        private static List<ContactInfo> contacts = new List<ContactInfo>
        {
            new ContactInfo { ContactId = 1, FirstName = "Sakshi", LastName = "Jadhav",
            CompanyName = "Cognizant", EmailId = "sakshi@gmail.com", MobileNo = 8765463534, Designation =
            "Software Engineer" },
            new ContactInfo { ContactId = 2, FirstName = "Mahi", LastName = "Reddy", CompanyName
            = "Capgemini", EmailId = "mah@gmail.com", MobileNo = 9876764337, Designation = "Frontend
            Developer" }
        };

        // SHOW ALL CONTACTS
        public ActionResult ShowContacts()
        {
            return View(contacts);
        }

        // GET CONTACT BY ID
        public ActionResult GetContactById(int id)
```

```

{
    var contact = contacts.FirstOrDefault(c => c.ContactId == id);
    if (contact == null)
    {
        return Content("Contact Not found!");
    }
    return View(contact);
}

// LOAD ADD CONTACT PAGE
public ActionResult AddContact()
{
    return View();
}

// SAVE CONTACT
[HttpPost]
[ValidateAntiForgeryToken]
public ActionResult AddContact(ContactInfo contact)
{
    if (ModelState.IsValid)
    {
        // Check if ID already exists (since we don't have a DB to handle unique keys)
        if (contacts.Any(c => c.ContactId == contact.ContactId))
        {
            ModelState.AddModelError("ContactId", "This ID is already taken!");
            return View(contact);
        }

        contacts.Add(contact);
        return RedirectToAction("ShowContacts");
    }
    return View(contact);
}
}
}

```

Views/ Contact/ AddContact.cshtml

@model ContactApp.Models.ContactInfo

<h2>Add New Contact</h2>

<div asp-validation-summary="ModelOnly" class="text-danger"></div>

<form asp-action="AddContact" method="post">

```

<div class="mb-3">
    <label asp-for="ContactId"></label>
    <input asp-for="ContactId" class="form-control" />
    <span asp-validation-for="ContactId" class="text-danger"></span>
</div>

```

```

<div class="mb-3">
    <label asp-for="FirstName"></label>
    <input asp-for="FirstName" class="form-control" />
    <span asp-validation-for="FirstName" class="text-danger"></span>
</div>

```

```

<div class="mb-3">
    <label asp-for="LastName"></label>
    <input asp-for="LastName" class="form-control" />
    <span asp-validation-for="LastName" class="text-danger"></span>
</div>

```

```

<div class="mb-3">
    <label asp-for="CompanyName"></label>
    <input asp-for="CompanyName" class="form-control" />
    <span asp-validation-for="CompanyName" class="text-danger"></span>
</div>

<div class="mb-3">
    <label asp-for="EmailId"></label>
    <input asp-for="EmailId" class="form-control" />
    <span asp-validation-for="EmailId" class="text-danger"></span>
</div>

<div class="mb-3">
    <label asp-for="MobileNo"></label>
    <input asp-for="MobileNo" class="form-control" />
    <span asp-validation-for="MobileNo" class="text-danger"></span>
</div>

<div class="mb-3">
    <label asp-for="Designation"></label>
    <input asp-for="Designation" class="form-control" />
    <span asp-validation-for="Designation" class="text-danger"></span>
</div>

<button type="submit" class="btn btn-primary">Save Contact</button>
</form>

@section Scripts {
    @{
        await Html.RenderPartialAsync("_ValidationScriptsPartial");
    }
}

```

Views/ Contact/ GetById.cshtml

```

@model ContactApp.Models.ContactInfo

<h2>Contact Details</h2>
<div class="card p-3">
    <p><strong>ID:</strong> @Model.ContactId</p>
    <p><strong>Name:</strong> @Model.FirstName @Model.LastName</p>
    <p><strong>Company:</strong> @Model.CompanyName</p>
    <p><strong>Email:</strong> @Model.EmailId</p>
</div>
<a asp-action="ShowContacts">Back to List</a>

```

Views/ Contact/ ShowContacts.cshtml

```

@model IEnumerable<ContactApp.Models.ContactInfo>

<h2 class="mt-3">Client Contact List</h2>

<div class="mb-3">
    <a asp-action="AddContact" class="btn btn-primary">Add New Contact</a>

    <button onclick="searchById()" class="btn btn-secondary">Search Contact</button>
</div>

<table class="table table-bordered table-striped">
    <thead class="table-dark">
        <tr>
            <th>ID</th>
            <th>First Name</th>

```

```

        <th>Last Name</th>
        <th>Company</th>
        <th>Email</th>
        <th>Mobile</th>
        <th>Designation</th>
        <th>Action</th>
    </tr>
</thead>
<tbody>
    @if (Model != null && Model.Any())
    {
        @foreach (var item in Model)
        {
            <tr>
                <td>@item.ContactId</td>
                <td>@item.FirstName</td>
                <td>@item.LastName</td>
                <td>@item.CompanyName</td>
                <td>@item.EmailId</td>
                <td>@item.MobileNo</td>
                <td>@item.Designation</td>
                <td>
                    <a asp-action="GetContactById" asp-route-id="@item.ContactId" class="btn
btn-sm btn-info">Details</a>
                </td>
            </tr>
        }
    }
    else
    {
        <tr>
            <td colspan="8" class="text-center text-muted">No contacts found in the
list.</td>
        </tr>
    }
</tbody>
</table>

@section Scripts {
    <script>
        function searchById() {
            // 1. Ask user for the ID
            var id = prompt("Enter Contact ID to search:");

            // 2. If user didn't hit cancel and entered a value
            if (id && id.trim() !== "") {
                // 3. Redirect to the GetContactById action in the Contact controller
                window.location.href = "/Contact/GetContactById/" + id;
            }
        }
    </script>
}

```

OUTPUT:

ContactApp Home Privacy

Client Contact List

Add New Contact

Search Contact

ID	First Name	Last Name	Company	Email	Mobile	Designation	Action
1	Sakshi	Jadhav	Cognizant	sakshi@gmail.com	8765463534	Software Engineer	Details
2	Mahi	Reddy	Capgemini	mahi@gmail.com	9876764337	Frontend Developer	Details
3	Kiran	Bedi	UPS	kiran@ups.com	7777755555	VP	Details

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ContactApp Home Privacy

Add New Contact

ContactId

Contact ID is required

FirstName

First Name cannot be empty

LastName

Last Name is required

CompanyName

The CompanyName field is required

EmailId

Email is required

MobileNo

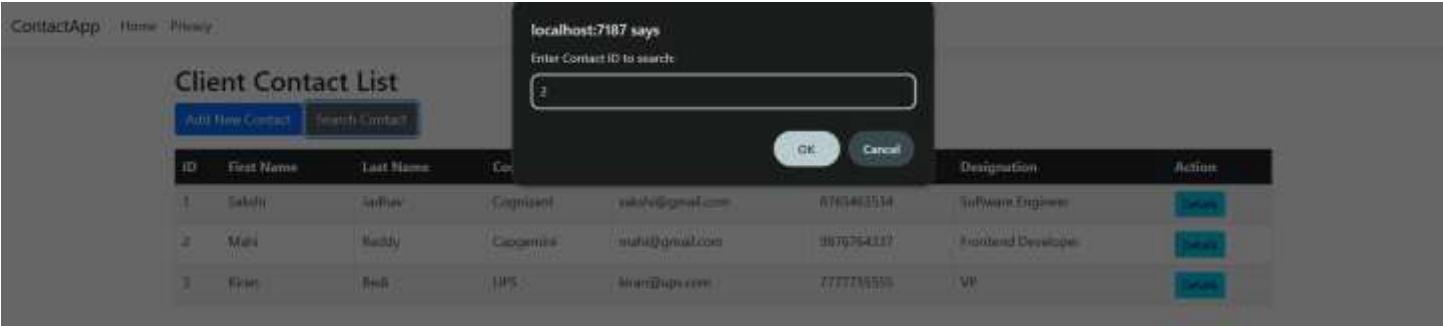
Mobile Number is required

Designation

The Designation field is required

Save Contact

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Contact Details

ID: 2

Name: Mahi Reddy

Company: Capgemini

Email: mahi@gmail.com

[Back to List](#)